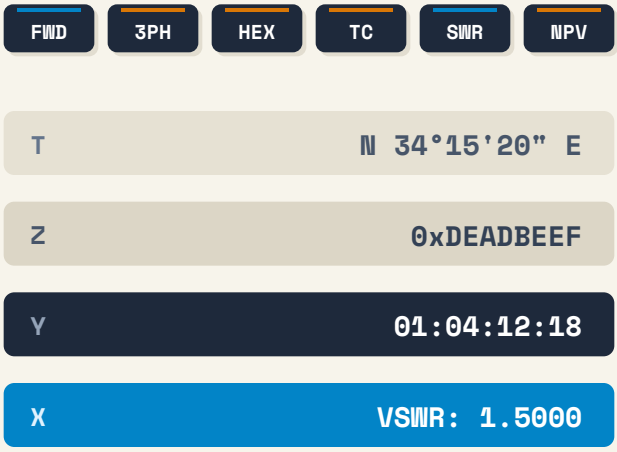
# THE STACK IN THE FIELD

*When Mathematics Meets the Physical Trades*




StackCalc

MATHEMATICS / APPLIED ENGINEERING & TRADES

# Every physical trade speaks its own dialect of mathematics.

Why do land surveyors measure angles clockwise from true North instead of counterclockwise from East? Why do bakers anchor every ingredient to flour weight rather than total dough weight? Why does radio power decay with the square of distance and frequency, and why must clinical drug doses scale with body surface area and renal clearance rather than body weight alone?

**The Stack in the Field** explores the first-principles mathematics behind nine physical disciplines—using compact operational acronyms (**FWD, INV, VD, BEND, HEX, TC, SWR, MW, BKR%, BSA, NPV**) to motivate each idea before showing how those equations unfold step by step across a four-register Reverse Polish Notation stack (**Registers T, Z, Y, X**) on the **SC-35, SC-14, SC-16, SC-24, SC-50, SC-60, SC-78, SC-80,** and **SC-95**.

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**Nine Disciplines Inside:** Geomatics & Flight Navigation • Electrical & Industrial Trades • Binary Logic & Embedded Timing • Cinematography, Timecode & Acoustics • RF Propagation & Link Budgets • Stoichiometry & Wet-Lab Analytics • Artisan Baking & Zymurgy • Clinical Dosing & Kinetics • Capital Valuation & Statistical Inference.

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